

Species-aware Module Review: tRNA m7G46 Methylation (UniPathway UPA00989) in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module / bucket:** unipathway:UPA00989 — "tRNA modification; N(7)-methylguanine-tRNA biosynthesis" **Candidate gene:** trmB (PP_5103 | UniProt Q88CS7)

Summary

The UniPathway UPA00989 module — SAM-dependent formation of **N7-methylguanosine at position 46 (m7G46)** in the tRNA variable loop (EC 2.1.1.33) — is a **single biochemical step catalyzed by a single enzyme**. In *Pseudomonas putida* KT2440 this module is **fully satisfied ("covered")** by one gene, **trmB / PP_5103** (UniProt **Q88CS7**, TRMB_PSEPK, a reviewed Swiss-Prot entry). The gene maps cleanly to the TrmB family (HAMAP MF_01057, TIGR00091, Pfam PF02390, InterPro IPR003358, PROSITE PS51625), and a proteome-wide search of UP000000556 confirms it is the **unique** EC 2.1.1.33 / PF02390 protein in KT2440. There is no paralog ambiguity, and no second subunit or accessory step is expected.

The target-strain annotation itself rests on **rule-based homology evidence** (UniRule/HAMAP MF_01057; protein existence "inferred from homology"; ECO:0000255) rather than a direct *P. putida* KT2440 experiment. However, **functional transfer is very strong**: KT2440 TrmB is **80.2% identical** across its full length to the experimentally validated *P. aeruginosa* PAO1 TrmB, which was directly shown to form m7G46 on 23 tRNA substrates and to modulate translation of Phe/Asp-rich mRNAs and H2O2 resistance ([PMID: 31428787](#)). Because both organisms are in the same genus and share the identical single-domain, single-subunit TrmB architecture, transfer of function to KT2440 is robust.

The single most important **curation action** is to flag the Cellular Component term **GO:0043527 "tRNA methyltransferase complex"** on Q88CS7 as a **likely over-propagated annotation**. That complex term derives from the eukaryotic m7G46 enzyme, an obligate **Trm8–Trm82 (METTL1–WDR4) heterodimer**, whereas bacterial TrmB is a **monomeric, stand-alone** class I

SAM-methyltransferase. The GO:0043527 annotation on Q88CS7 carries only phylogenetically propagated (IEA:TreeGrafter) evidence and should be removed or down-weighted for the bacterial ortholog. With that caveat, the module is clean: one step, one enzyme, high-confidence covered.

Target-organism pathway definition

Process included (and only this): UPA00989 covers the transfer of a methyl group from S-adenosyl-L-methionine (SAM) to the **N7 atom of guanine 46** in the variable/tertiary core of cytoplasmic tRNAs, producing **m7G46 + S-adenosyl-L-homocysteine (SAH)**. m7G46 stabilizes tRNA tertiary structure and, in bacteria, feeds a "modification network" that promotes other modifications and tRNA stability ([PMID: 19934251](#)).

Reaction (UniProt catalytic activity):

guanosine(46) in tRNA + SAM --TrmB--> N7-methylguanosine(46) in tRNA + SAH (EC 2.1.1.33)

Enzyme: TrmB (PP_5103, Q88CS7), a 240-aa class I (Rossmann-fold) SAM-dependent methyltransferase.

Alternate names / database-specific definitions: - **UniPathway:** UPA00989 — "tRNA modification; N(7)-methylguanine-tRNA biosynthesis." - **Enzyme:** EC **2.1.1.33**; names "tRNA (guanine-N(7)-)-methyltransferase," "tRNA (guanine(46)-N(7))-methyltransferase," "tRNA(m7G46)-methyltransferase." - **Gene name:** *trmB* (bacterial; = *yggH* in *E. coli*). Eukaryotic orthologs are **Trm8/Trm82** (yeast) and **METTL1/WDR4** (human) — the same chemistry, but a *different family boundary and quaternary structure* that must **not** be merged into the bacterial module. - **Family identifiers:** HAMAP **MF_01057**; TIGRFAM **TIGR00091**; InterPro **IPR003358**; Pfam **PF02390** (Methyltransf_4); PROSITE **PS51625**; SUPFAM **SSF53335**.

Boundaries — what to keep separate: - This is **not** a general "RNA methylation" or "methyltransferase complex" bucket. - Other tRNA methyltransferases (TrmA/m5U54, TrmD/m1G37, TrmH/Gm18, TrmFO, m5C, etc.) are **distinct modules** that share only the class I SAM-MTase fold; do not collapse them into UPA00989. - Overview/superpathway "tRNA modification" maps should be kept separate from this single-step module.

Expected step model

UPA00989 is a **one-step module**:

Step	Reaction	Expected enzyme	KT2440 gene	Status
1 (only step)	tRNA-G46 + SAM → tRNA-m7G46 + SAH	tRNA (guanine46-N7)-methyltransferase (EC 2.1.1.33)	trmB / PP_5103 (Q88CS7)	Covered

There are no upstream/downstream enzymatic steps within this UniPathway. **No accessory/second subunit is expected** — the eukaryotic Trm82/WDR4 partner has no bacterial counterpart, so a "missing second subunit" is `not_expected_in_target_taxon`, not a gap. SAM supply and tRNA substrate provision are shared cellular resources, not module steps.

Key Findings

Finding 1 — The module is satisfied by a single gene, trmB/PP_5103 (Q88CS7)

The UPA00989 module is fully satisfied in *P. putida* KT2440 by a single gene. UniProt **Q88CS7** (TRMB_PSEPK, reviewed Swiss-Prot) is annotated with gene *trmB*, ordered locus **PP_5103**, EC 2.1.1.33, recommended name "tRNA (guanine-N(7)-)-methyltransferase," and a PATHWAY annotation of "tRNA modification; N(7)-methylguanine-tRNA biosynthesis" — exactly UPA00989. The catalytic activity is defined as guanosine(46) in tRNA + SAM → m7G46 in tRNA + SAH. The protein is 240 aa with a class I SAM-MTase fold, and carries a consistent stack of family assignments (HAMAP MF_01057, TIGR00091, InterPro IPR003358, Pfam PF02390, PROSITE PS51625). This is an unambiguous, reviewed, single-step-to-single-gene mapping.

Finding 2 — TrmB function is experimentally validated in the same genus

The functional assignment is anchored by direct experiments in related bacteria. In *P. aeruginosa*, *trmB* was shown to encode a tRNA guanine(46)-N7-methyltransferase that forms m7G46 in the tRNA variable loop, acting on 23 identified tRNA substrates; loss of *trmB* reduced translation of Phe/Asp-enriched mRNAs (*kata*, *katB*) and caused H2O2 sensitivity (PMID: 31428787). In *Thermus thermophilus*, TrmB/m7G46 is required for viability at high temperature via a tRNA modification network (PMID: 19934251). For KT2440 itself, the UniProt evidence is

rule-based (ECO:0000255; UniRule MF_01057; protein existence "inferred from homology"), so there is no direct *P. putida* experiment — but the same-genus *Pseudomonas* validation makes transfer strong.

[PMID: 31428787]: "the *P. aeruginosa* trmB gene encodes a tRNA guanine (46)-N7-methyltransferase that catalyzes the formation of m7G46 in the tRNA variable loop"

[PMID: 19934251]: "N(7)-methylguanine at position 46 (m(7)G46) in tRNA is produced by tRNA (m(7)G46) methyltransferase (TrmB)"

Finding 3 — GO:0043527 "tRNA methyltransferase complex" is a likely over-propagation

Q88CS7 carries **GO:0043527** (C: tRNA methyltransferase complex) with evidence code **IEA:TreeGraft** — purely phylogenetic propagation, not experimental — alongside the appropriate GO:0008176 (F: tRNA guanine46-N7-methyltransferase activity, IEA:UniProtKB-UniRule). The complex term is a **eukaryotic feature**: the yeast m7G46 enzyme is an obligate **Trm8–Trm82 heterodimer** (PMID: 18029735), and the human ortholog is the METTL1–WDR4 heterodimer (PMID: 36599982). By contrast, bacterial TrmB acts as a **single subunit** with less strict tRNA recognition, and a structural model describes it as a stand-alone class I SAM-MTase bound to tRNA (PMID: 34006170). The heterogeneous PANTHER family PTHR23417 to which this entry maps ("KDO transferase / tRNA guanine-N7-methyltransferase," garbled subfamily label) is the likely source of the noisy complex term.

[PMID: 18029735]: "Yeast tRNA (m(7)G46) methyltransferase contains two protein subunits (Trm8 and Trm82)"

Finding 4 — PP_5103 is the unique TrmB/EC 2.1.1.33 protein in KT2440 (no paralog, no missing subunit)

A UniProtKB proteome search of UP000000556 returns **exactly one** EC 2.1.1.33 hit (Q88CS7/PP_5103), **exactly one** Pfam PF02390 (Methyltransf_4/TrmB) hit (Q88CS7), and the gene name *trmB* returns only Q88CS7. There is no second TrmB-family paralog and no annotated Trm82/WDR4-like accessory-subunit gene. This confirms a clean one-step-to-one-gene mapping with no paralog ambiguity and no genuine subunit gap.

Finding 5 — 80% identity to the validated *P. aeruginosa* TrmB makes transfer very strong

Global (Needleman–Wunsch) pairwise alignment gives: Q88CS7 (KT2440, 240 aa) vs Q9I6B3 (TRMB_PSEAE, *P. aeruginosa* PAO1, 244 aa — the enzyme characterized in

P 31428787

= **80.2% identity** over 237 aligned residues; vs P0A8I5 (TRMB_ECOLI, 239 aa) = 59.4%. All three are full-length single-domain TrmB orthologs. Such high full-length identity to a directly assayed same-genus ortholog upgrades confidence in the "covered" call well beyond generic UniRule homology.

Comparison	Identity	Aligned residues	Basis
KT2440 Q88CS7 vs <i>P. aeruginosa</i> Q9I6B3	80.2%	237	Same-genus, experimentally validated ortholog
KT2440 Q88CS7 vs <i>E. coli</i> P0A8I5	59.4%	full-length	Distant bacterial ortholog

Mechanistic Model / Interpretation

The module is best understood as a **single node** with well-defined inputs/outputs and an experimentally anchored genus-level ortholog:

```

      SAM                      SAH
      |                        ^
      v                        |
[ tRNA-G46 ] --- TrmB (PP_5103, Q88CS7, MONOMER) ---> [ tRNA-m7G46 ]
                  |
            80.2% identity to P. aeruginosa TrmB
            (Q9I6B3, experimentally validated, <a href="https://pubmed.ncbi.nlm.nih.gov/31428787">https://pubmed.ncbi.nlm.nih.gov/31428787</a>)
                  |
            59.4% identity to E. coli TrmB (P0A8I5)

Eukaryotic contrast: Trm8–Trm82 (yeast) / METTL1–WDR4 (human) = obligate HETERODIMER
Bacterial reality:   TrmB = single-subunit enzyme => GO:0043527 (complex) is OVER-PROPAGATED

```

The curation narrative is simple and confident: (1) the module is one step; (2) KT2440 encodes exactly one enzyme for it; (3) that enzyme is nearly identical to a directly validated *Pseudomonas* ortholog; (4) the only quaternary-structure annotation in the record (a complex CC term) is a eukaryote-derived artifact and should be flagged. Downstream biology — the tRNA-modification network, rapid tRNA decay, and translational/oxidative-stress phenotypes — is a *consequence* of the modification, not an additional catalytic step, and must not expand the module boundary.

Candidate gene assessment (curation detail)

trmB — PP_5103 — Q88CS7 (TRMB_PSEPK)

Attribute	Value
Role	tRNA (guanine46-N7)-methyltransferase; forms m7G46 (GO:0008176)
Target-strain evidence	Rule-based homology (ECO:0000255; HAMAP MF_01057; protein existence "inferred from homology"); reviewed Swiss-Prot
Family support	HAMAP MF_01057; TIGR00091; Pfam PF02390; InterPro IPR003358; PROSITE PS51625; SUPFAM SSF53335
Length	240 aa (consistent with monomeric bacterial TrmB)
Cross-species support	<i>P. aeruginosa</i> (P 31428787), <i>T. thermophilus</i> (P 19934251)
Sequence transfer	80.2% identical to validated <i>P. aeruginosa</i> Q9I6B3; 59.4% to <i>E. coli</i> P0A8I5
Paralog ambiguity	None — sole EC 2.1.1.33/PF02390/ <i>trmB</i> protein in UP000000556

Curation caveats: - **GO:0043527** "tRNA methyltransferase complex" (IEA:TreeGrafter) is over-propagated from the eukaryotic heterodimer; flag/remove for the monomeric bacterial entry. - **PANTHER PTHR23417** is heterogeneous (mixes KDO transferase / tRNA-guanine-N7-MTase); do

not use PANTHER subfamily text as the authoritative name — HAMAP/TIGRFAM are the reliable anchors. - Broad fold-level EC/GO terms (class I SAM-MTase) are shared with unrelated MTases, but the specific TrmB signatures resolve identity cleanly.

Gaps, ambiguities, and likely over-annotations

- **No missing steps.** The pathway has one reaction and it is encoded.
- **Proteome-wide uniqueness (verified):** exactly one EC 2.1.1.33, one PF02390, and one *trmB* in UP000000556 — no second paralog, no annotated Trm82/WDR4-like subunit.
- **Not expected in this organism:** a Trm82/WDR4-like accessory subunit (contrast the yeast obligate heterodimer, [PMID: 18029735](#)). Do **not** open a gap for a missing second subunit.
- **Likely over-annotation:** GO:0043527 "tRNA methyltransferase complex" (CC) — over-propagated; the MF term GO:0008176 is correct.
- **Direct target-strain data gap:** no published *P. putida* KT2440 biochemical/knockout study of *trmB*; the annotation rests on UniRule homology plus strong genus-level experiments.

Module and GO-curation recommendations

Module element	Recommended status	Rationale
Step 1: m7G46 formation (trmB/PP_5103)	covered (high confidence)	Sole EC 2.1.1.33/PF02390 protein; 80% identical to validated <i>P. aeruginosa</i> TrmB
Second subunit / accessory step	not_expected_in_target_taxon	Bacterial TrmB is monomeric; no Trm82/WDR4 counterpart
Overall module UPA00989	satisfied / covered	Single-step, single-enzyme; no residual gaps
Generic module boundaries	correct for this organism	One-step scope matches bacterial biology; no revision needed

GO-curation actions: 1. **Flag/remove** **GO:0043527** (tRNA methyltransferase complex) on Q88CS7 as over-propagated for the monomeric bacterial enzyme (evidence IEA:TreeGrafter only). 2. **Retain** GO:0008176 (tRNA guanine46-N7-methyltransferase activity) and a tRNA-methylation BP term (e.g., GO:0106004 / GO:0030488). 3. **No new module document or GO term is needed** — existing terms cover the biology; the only action is trimming the complex CC term.

Genes to promote to full review

- **trmB (PP_5103 / Q88CS7):** promote to full `fetch-gene` review **only to record the GO complex-term correction** and to upgrade the evidence note (cite same-genus experimental validation,

P 31428787

record 80% identity to Q9I6B3). The gene identity is unambiguous and does not require reassignment. Priority: **low-to-moderate** (annotation cleanup, not identity resolution). No other genes warrant promotion, as the proteome contains no paralog or candidate alternative.

Evidence Base

PMID	Paper (abbrev.)	Role in this review
31428787	<i>TrmB, a tRNA m7G46 methyltransferase ... in Pseudomonas aeruginosa</i> (2019)	Same-genus experimental anchor. Direct demonstration that <i>Pseudomonas</i> TrmB forms m7G46 (23 substrates) and affects katA/katB translation and H2O2 resistance — the strongest transfer basis for KT2440.
19934251	<i>m7G46 in tRNA from Thermus thermophilus required for viability at high temperature</i> (2010)	Confirms TrmB catalyzes the defining UPA00989 reaction and places m7G46 in a tRNA modification network.
18029735	<i>Hetero subunit interaction ... of yeast tRNA (m7G46) methyltransferase</i> (2007)	Establishes the complex/heterodimer as a eukaryotic feature (Trm8+Trm82) — the basis for flagging GO:0043527 in the bacterial entry.
34006170	<i>Structural model of the M7G46 methyltransferase TrmB in complex with tRNA</i> (2021)	Describes bacterial TrmB as a monomeric, stand-alone class I SAM-MTase.
36599982	<i>Structures and mechanisms of tRNA methylation by METTL1-WDR4</i> (2023)	Human ortholog is a heterodimer — parallels the yeast case, contrasts with bacteria.
16387656	<i>Rapid tRNA decay from lack of nonessential modifications</i> (2006)	Places m7G46 (Trm8/Trm82) in the RTD network — downstream biology, out of module scope.
22895820	<i>Yeast RTD pathway competes with EF-1A for hypomodified tRNAs</i> (2012)	Further RTD/network context; not part of the catalytic module.
38977661	<i>Perturbation of METTL1-mediated tRNA m7G ...</i> (2024)	Eukaryotic functional consequences of m7G46 loss; context only.

Additional anchors: UniProt **Q88CS7** (TRMB_PSEPK); HAMAP **MF_01057**; UniPathway **UPA00989**.

Limitations and Knowledge Gaps

- **No direct KT2440 experiment.** All target-strain evidence is homology/UniRule (ECO:0000255). The functional call depends on transfer from *P. aeruginosa*; while 80% single-domain identity makes this strong, it is not a KT2440-specific demonstration of m7G46 formation.
 - **Substrate scope inferred.** The 23-tRNA substrate set was mapped in *P. aeruginosa*; the exact set of KT2440 tRNAs bearing m7G46 is not experimentally defined.
 - **Phenotype unverified in KT2440.** The H₂O₂-resistance / catalase-translation role is a *P. aeruginosa* finding; whether *trmB* loss produces the same oxidative-stress phenotype in KT2440 is untested.
 - **PANTHER family noise.** The heterogeneous PTHR23417 assignment is the probable origin of the over-propagated complex term; other spuriously propagated terms should be checked during full review.
-

Proposed Follow-up Experiments / Actions

1. **Curation (immediate):** Flag/remove **GO:0043527** on Q88CS7; retain GO:0008176 and a tRNA-methylation BP term. Record the *P. aeruginosa* 80%-identity ortholog and

P 31428787

 as the transfer basis in the gene note.
 2. **Mark module UPA00989 as covered** for KT2440 with a single-step model; record "second subunit" as `not_expected_in_target_taxon`.
 3. **Optional experimental confirmation (low priority):** LC-MS/MS or primer-extension analysis of bulk KT2440 tRNA $\pm \Delta trmB$ (PP_5103) to directly confirm m7G46 in the target strain and enumerate KT2440 substrates.
 4. **Optional phenotype test:** Assess H₂O₂ sensitivity of a KT2440 $\Delta trmB$ mutant to test whether the *P. aeruginosa* oxidative-stress role transfers, given KT2440's own catalase repertoire.
 5. **No new module document or GO term request is needed.**
-

Uncertainty statement

The only *direct* target-strain evidence is homology-based (UniProt ECO:0000255, UniRule). All functional claims are inferred from (a) unambiguous TrmB family signatures, (b) 80.2% full-length identity to the experimentally validated *P. aeruginosa* PAO1 TrmB (Q9I6B3), and (c) experimental data in related bacteria (*P. aeruginosa*, *T. thermophilus*). Transfer of function to *P. putida* KT2440 is judged **strong**; the complex-term over-annotation judgment is **high confidence**, based on the well-established monomeric nature of bacterial TrmB and the proteome-wide absence of any second subunit.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery